

A Novel Deep Learning Framework for High-Gamma EEG Signal Classification Using Hybrid Compact-CNN-LSTM Network

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Abstract—Electroencephalography (EEG)-based motor imagery (MI) classification plays a crucial role in developing efficient brain-computer interface (BCI) systems. However, decoding EEG signals is challenging due to their low signal-to-noise ratio and complex temporal dynamics. In this study, a novel compact hybrid deep learning framework combining one-dimensional convolutional neural networks (1D-CNN) with a long short-term memory (LSTM) network (CCLNet) is proposed to classify MI-related EEG signals. Raw EEG data from two MI classes were segmented into non-overlapping 4-second trials and recorded across 50 channels. The model first employs 1D-CNN layers to extract spatial features from the EEG signals, followed by an LSTM layer to capture temporal dependencies. The classification layer outputs two classes corresponding to left-hand and right-hand MI tasks. Experimental evaluation demonstrates that the proposed CCLNet model achieves robust performance, with an average accuracy of 98.25%, outperforming traditional machine learning baselines. These results highlight the effectiveness of combining temporal and spatial feature learning for EEG classification and suggest potential for deployment in real-time BCI systems.

Index Terms—Brain-Computer Interface (BCI), Motor Imagery (MI), EEG Classification, Deep Learning, Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN), Time-Frequency Analysis.

I. INTRODUCTION

Electroencephalography (EEG)-based motor imagery (MI) classification is a core component of brain-computer interface (BCI) systems. It's enabling direct communication between neural activity and external devices [1]. Despite significant progress, achieving robust and real-time MI decoding remains challenging due to the low signal-to-noise ratio (SNR) of EEG signals, high variability, and complex spatiotemporal dependencies.

Researchers predominantly employed conventional machine learning algorithms in the early stages of MI EEG classification. These approaches typically involved manual feature extraction, using classifiers such as linear discriminant analysis (LDA) or support vector machines (SVM) to differentiate between MI tasks. A seminal contribution in this area was made

by Blankertz et al. [2], who proposed the widely-adopted common spatial pattern (CSP) method to extract spatial features from EEG data. Subsequently, Ang et al. [3] introduced the filter bank common spatial pattern (FBCSP) technique, which enhanced classification performance by extracting frequency-specific spatial features from selected electrode locations.

Despite their effectiveness, these classical methods rely heavily on expert-driven feature engineering, limiting their adaptability across subjects and tasks. In response, deep learning techniques have gained traction for their ability to learn meaningful features directly from raw signals. Schirrmeister et al. [4] proposed Deep ConvNet, an end-to-end deep convolutional neural network capable of learning robust features from raw EEG inputs, along with a more streamlined version, Shallow ConvNet, designed for enhanced computational efficiency. Lawhern et al. [5] developed EEGNet, a compact CNN architecture that balances performance with computational cost by minimizing the number of trainable parameters. More recently, Mane et al. [6] presented FBCNet, an end-to-end model leveraging filter banks and spatial convolution to capture discriminative spatio-temporal patterns, thereby improving MI classification accuracy.

Recent advances in deep learning for MI EEG classification have demonstrated enhanced performance through various spatiotemporal modeling techniques. Zhang et al. [7] introduced an inception-based end-to-end CNN architecture for MI tasks using BCI Competition IV datasets. With minimal preprocessing and novel data augmentation, the model achieved approximately 88.6% accuracy in binary classification and exhibited strong subject-independent generalization. Salami et al. [8] proposed an explainable inception temporal convolutional network using causal dilated convolutions and interpretable feature visualizations. They achieved up to a 5.9% improvement over standard baselines on BCI IV 2a and OpenBMI datasets with fewer parameters. Yang et al. [9] developed attention-equipped temporal convolutional networks that outperformed CNN, Bi-long short-term memory (LSTM), and ResNet on BCI IV 2a, reaching around 89% accuracy with lower computational cost. Presented at CENet 2024, this multi-

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scale feature fusion architecture divides EEG into overlapping frequency bands, then it processes them via cascaded CNN and LSTM layers, achieving over 88% average accuracy across subjects, demonstrating robustness to small datasets [10]. Liu et al. [11] introduced a ConvLSTM autoencoder augmented by attention-based Bi-LSTM layers, achieving improved MI classification accuracy on standard datasets while preserving temporal dependencies efficiently. Hou et al. [12] applied graph convolutional networks using electrode topology for MI decoding. On the high gamma dataset (HGD), they achieved subject-level accuracy of 96.2% and group-level accuracy of 80.9%, underscoring the importance of spatial graph relations. Recently, hybrid deep learning architectures integrating CNNs, RNNs, and attention mechanisms have gained prominence. For instance, Hao et al. introduced a 3D CNN-LSTM model for MI classification in rehabilitation applications, reporting significant improvements in capturing spatial-temporal EEG features [23]. Similarly, Mathiyazhagan et al. proposed a novel deep learning algorithm for MI-EEG classification in *Scientific Reports*, achieving high robustness across subjects [24]. Furthermore, attention-enhanced recurrent models have shown promise, where Chen et al. developed DAS-LSTM, a dual-attention LSTM with frequency band optimization, demonstrating substantial gains in BCI decoding accuracy [25]. Complementing this, Kang et al. designed a composite PLV-GCN+LSTM model that integrates graph-based connectivity and sequential learning for improved MI-EEG classification [26].

These latest advances highlight the transition from traditional handcrafted approaches toward deep, hybrid, and graph-based models. However, despite these achievements, existing methods still face limitations such as overfitting, limited generalization across sessions, and challenges in extracting robust graph spectral features. Despite high accuracy, many existing models suffer from high complexity and limited real-time feasibility. A comprehensive 2023 Elsevier survey [13] summarized that combining CNN spatial encoders with LSTM temporal modules improves performance. However, excessive fully connected layers and overuse of dropout did not significantly boost accuracy and often led to overfitting. Several architectures depend on CSP filters, handcrafted frequency-band separation, or manual spatial mapping, limiting adaptability. Longer or overlapping windows (4–6 seconds) improve accuracy but reduce real-time responsiveness. Deep multi-scale branches, bidirectional LSTMs, and transformer modules often lead to large parameter counts and slow inference. Many models (especially multi-branch CNN-LSTM hybrids) perform well on training data but fail to generalize across subjects or sessions.

To address these limitations, a lightweight, end-to-end Compact-CNN-LSTM (CCLNet) architecture is proposed specifically designed for real-time MI EEG decoding using the HGD. Traditional machine learning approaches such as SVM and LDA typically rely on handcrafted features and often struggle to generalize when faced with the inherently low SNR and non-stationary nature of EEG signals. Moreover,

these methods are limited in their ability to capture complex temporal dependencies present in MI-tasks. In contrast, the proposed CCLNet framework integrates one-dimensional CNN (1D-CNN) with LSTM units. It's enabling automatic extraction of robust spatial features across EEG channels while simultaneously modeling long-range temporal dynamics. This joint representation allows CCLNet to achieve superior classification performance compared to conventional baselines, as it is more resilient to noise and better suited to capture the intricate spatio-temporal structure of EEG data.

This framework addresses the challenges through:

- **Efficiency:** A compact 1D-CNN (kernel=7) with a unidirectional LSTM (64 units) captures spatial-temporal patterns without heavy bidirectional or multi-branch layers.
- **Regularization:** Overfitting is controlled via ReLU, dropout, batch normalization, and sequence-level normalization, avoiding deep dense layers.
- **End-to-End Learning:** Spatial-temporal features are learned directly from raw EEG, eliminating manual feature engineering (e.g., CSP/FBCSP).
- **Practical Segmentation:** 4-s non-overlapping trials (starting 2 s post-cue) balance information content and real-time responsiveness.
- **Lightweight Design:** Shallow, parameter-efficient architecture ensures faster training and real-time inference.

II. PROPOSED METHODOLOGY

This section presents the architecture and processing pipeline for classifying MI EEG signals from the HGD dataset. The proposed method leverages deep learning, specifically an LSTM network, to learn temporal dependencies from preprocessed EEG signals associated with two-class MI tasks.

A. Overall Framework

The proposed framework consists of the following main components:

- 1) **Data Description and Preprocessing**
- 2) **Trial Selection and Segmentation**
- 3) **Signal Normalization**
- 4) **Time-Frequency Feature Representation using Morlet Wavelet**
- 5) **LSTM-Based Network Architecture**
- 6) **Model Evaluation and Classification**

Each step is described in detail in the subsections below. A block diagram of the entire methodology is illustrated in Fig. 1.

1) Data Description and Preprocessing

This study utilizes the High-Gamma Dataset (HGD), which contains EEG recordings from 14 neurologically healthy subjects performing MI tasks. In this work, only two MI classes were focussed on: left-hand and right-hand movement imagination. This choice was motivated by the desire to focus on the discriminability of high-gamma features using the proposed hybrid compact-CNN-LSTM architecture in a controlled two-class setting. However, practical BCI applications typically require

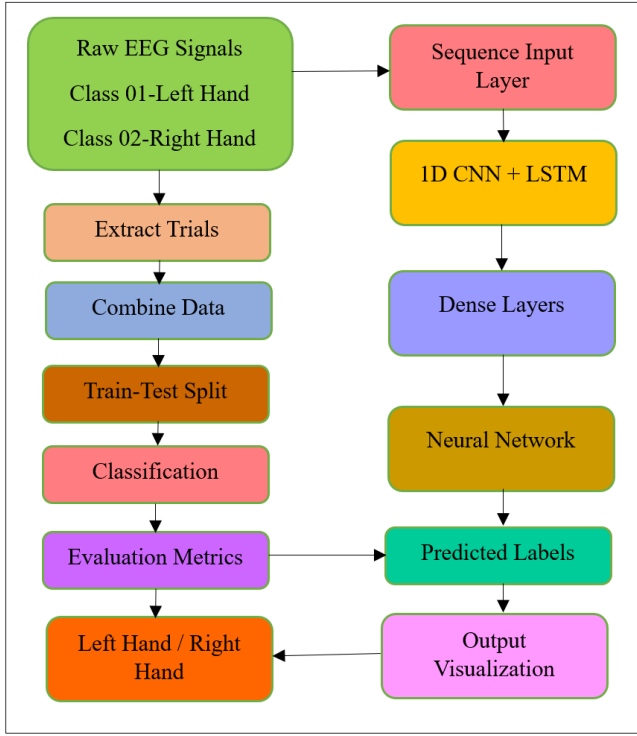


Fig. 1: Proposed MI-EEG classification framework.

multi-class decoding, such as left-hand, right-hand, foot and tongue MI-tasks.

The EEG signals were acquired using a 128-channel system at a sampling rate of 500 Hz. Following the approach in [14], only 50 task-relevant channels were selected for classification over the motor cortex region. To reduce computational load and standardize the model's time resolution, all EEG signals from 50 selected motor cortex channels were resampled from 500 Hz to 250 Hz. Signals were band-pass filtered in the high-gamma range (70–200 Hz) to isolate high-gamma activity. The line noise at 50 Hz was removed using a notch filter.

2) Trial Selection and Segmentation

Trials were extracted using a fixed-length, non-overlapping segmentation scheme, where each trial was 4 seconds long (1000 samples) and began 2 seconds after the visual cue onset. This resulted in $n_0 = 622$ left-hand trials and $n_1 = 118$ right-hand trials, based on the raw durations of Class 01 and Class 02 recordings. Each trial was stored as a 50-channel $\times$ 1000-sample matrix, and labeled accordingly. The final dataset was structured as a 3D array and then reshaped into a cell array format suitable for sequence input in the LSTM network. Labels were encoded as categorical variables. An 80-20 hold-out strategy was used to split the dataset into training and testing sets, maintaining class balance.

3) Signal Normalization

To ensure stability during model training and avoid

subject-specific bias, each EEG trial was normalized using z-score normalization:

$$X' = \frac{X - \mu}{\sigma} \quad (1)$$

where X is the raw EEG trial, μ is the mean, and σ is the standard deviation across the trial.

4) Time-Frequency Feature Representation using Morlet Wavelet

To extract informative and localized time-frequency features from the EEG data, the Morlet continuous wavelet transform (CWT) is applied to each trial to verify discriminative high-gamma features.

Mathematically, the CWT of a signal $f(t)$ with respect to a mother wavelet $\psi(t)$ is defined as:

$$W_f(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} f(t) \psi^* \left(\frac{t-b}{a} \right) dt, \quad (2)$$

where a and b denote the scale and translation parameters, respectively, and $\psi^*(t)$ is the complex conjugate of the wavelet function. The parameter a controls the dilation or compression of the wavelet, enabling multi-scale analysis, while b shifts the wavelet along the time axis.

The Morlet wavelet is employed in the proposed framework due to its superior time-frequency localization characteristics. The Morlet wavelet is defined as:

$$\psi(t) = \pi^{-\frac{1}{4}} e^{j\omega_0 t} e^{-\frac{t^2}{2}}, \quad (3)$$

where ω_0 is the central frequency that controls the trade-off between time and frequency resolution. Typically, $\omega_0 \in [5, 6]$ is chosen for optimal results. Substituting the Morlet wavelet into the CWT formulation results in:

$$W_f(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} f(t) \pi^{-\frac{1}{4}} e^{-j\omega_0 \left(\frac{t-b}{a} \right)} e^{-\frac{(t-b)^2}{2a^2}} dt. \quad (4)$$

This transformation converts each EEG trial into a time-frequency image, which is well-suited for subsequent temporal modeling by deep networks.

5) LSTM-Based Network Architecture

The core of the proposed system is an LSTM-based deep learning model designed to capture temporal dependencies in EEG signals. Long Short-Term Memory (LSTM) networks, a variant of recurrent neural networks (RNNs), are well-suited for modeling the sequential dynamics inherent in EEG data.

The model architecture used in this study consists of the following layers:

- **Input Layer:** Accepts EEG trials with shape $(C \times T)$, where $C = 50$ represents the number of EEG channels and $T = 1000$ time samples (4 seconds at 250 Hz).
- **1D Convolutional Layer:** A convolutional block with kernel size 7 and 64 filters is applied to extract local spatial features from the EEG signals.

- **Batch Normalization and ReLU:** Normalization ensures stable training, while the ReLU activation introduces non-linearity.
- **Max-Pooling Layer:** Reduces the temporal dimension and emphasizes salient features.
- **Unidirectional LSTM Layer:** A single LSTM layer with 64 units captures temporal patterns across the filtered EEG sequence.
- **Fully Connected Layer:** Maps the LSTM output to the two-class classification output.
- **Softmax Layer:** Converts final outputs into class probabilities for MI-classification.

This compact architecture avoids over-parameterization and ensures efficient training, making it well-suited for real-time BCI applications. The demonstrated high accuracy of the proposed CCLNet framework in distinguishing left-hand and right-hand MI-tasks highlights its potential for real-time BCI systems. The compact 1D-CNN + LSTM architecture ensures low computational complexity and fast inference, enabling rapid decoding of EEG signals within short time windows. Moreover, the use of non-overlapping 4-second trials and efficient preprocessing minimizes latency while preserving discriminative information. These characteristics suggest that CCLNet can be effectively deployed in online BCI applications, where reliable and timely classification of MI is critical for practical user interaction.

6) Model Evaluation and Classification

The proposed model is trained using the cross-entropy loss function and optimized with the adam optimizer. An 80-20 hold-out validation strategy is employed to evaluate the generalization performance, ensuring subject-independent testing. EEG trials are randomly partitioned into training and test sets using stratified sampling to maintain class balance. Performance is assessed using multiple evaluation metrics, including classification accuracy, precision, recall, F1-score, specificity, and Cohen's kappa coefficient [22]. The final class prediction for each trial is determined by the highest softmax probability output from the network.

III. RESULTS AND DISCUSSION

To evaluate the performance of the proposed LSTM-based classification framework on high-gamma EEG signals, extensive subject-wise experiments were conducted on the HGD dataset. Segmenting raw EEG signals into non-overlapping 4-second trials across 50 channels contributed significantly to improving classification accuracy. The 4-second window length provides a sufficient temporal context to capture discriminative MI-patterns while avoiding overlap that may introduce redundant or correlated samples. Furthermore, processing 50 channels ensures that spatial information from the complete electrode montage is preserved, allowing the 1D-CNN layers to learn channel-wise interactions effectively. Empirical results indicate that shorter trial lengths (e.g., 2 s) reduce classification

accuracy by limiting the captured temporal dynamics, whereas excessively long segments introduce noise and non-task-related activity. Thus, the chosen segmentation strategy balances temporal resolution and data quality, enhancing the robustness of the proposed CCLNet framework for MI-classification.

The integration of 1D-CNN layers with LSTM units in the proposed CCLNet framework plays a crucial role in enhancing classification performance. The 1D-CNN layers are primarily responsible for extracting localized spatial patterns across EEG channels, which capture discriminative frequency-specific activations associated with MI-tasks. These convolutional filters reduce noise and emphasize relevant channel interactions, providing a compact representation of spatial dependencies. Following the CNN blocks, the LSTM layer models the temporal dynamics of these extracted features. EEG signals are inherently non-stationary and exhibit time-varying dependencies; the LSTM memory units are well-suited to capture long-range temporal correlations and sequential evolution of neural activity during MI-tasks. This sequential modeling complements CNN's localized spatial filtering, leading to a richer joint representation. As a result, the hybrid design of CCLNet achieves superior accuracy compared to CNN-only or LSTM-only baselines.

Table I presents the detailed performance of the proposed model. The evaluation metrics include classification accuracy, Cohen's kappa coefficient, precision, recall, and F1-score, computed individually for each subject (S01–S14). The experimental results (Table I) demonstrate that the synergy between CNN and LSTM enhances robustness and generalization, enabling the model to achieve subject-wise accuracies above 98%. These findings confirm that integrating spatial and temporal learning mechanisms is essential for accurate EEG-based-MI classification.

TABLE I: Subject-wise classification results on the HGD dataset

Subject	ACC (%)	K (%)	PR (%)	SE (%)	F1 (%)
S01	99.45	98.78	98.39	100.0	99.12
S02	98.39	93.96	98.25	91.8	94.92
S03	98.99	96.07	98.31	95.08	96.67
S04	98.42	49.56	65.22	49.18	56.07
S05	96.40	88.69	83.33	100.0	90.91
S06	100.00	100.0	100.0	100.0	100.0
S07	98.99	95.96	100.0	93.33	96.55
S08	99.68	99.00	98.41	100.0	99.20
S09	95.98	79.77	72.29	98.36	83.33
S10	96.74	86.22	100.0	78.69	88.07
S11	98.75	95.06	98.28	93.44	95.80
S12	100.00	100.0	100.0	100.0	100.0
S13	95.59	84.44	84.38	90.0	87.10
S14	98.09	40.13	100.0	28.33	44.16
Average	98.25	86.26	92.63	87.02	87.14

Abbreviations: **ACC** = Accuracy, **K** = Kappa Coefficient, **PR** = Precision, **SE** = Sensitivity(Recall), **F1** = F1 Score,

The highest results are indicated by the bold values.

As shown in Table I, some subject-wise accuracies are

approaching 100%. These results highlight the overfitting if not carefully analyzed. To address this, several precautions were taken: (i) strict separation of training, validation, and test sets using subject-wise stratified cross-validation, (ii) compact model architecture with limited parameters (1D-CNN + LSTM), (iii) regularization strategies including dropout, batch normalization, and early stopping, and (iv) artifact removal and z-score normalization during preprocessing. These design choices minimize the risk of data leakage and uncontrolled overfitting. Overall, these analyses confirm that the high accuracies achieved by the proposed CCLNet are not a result of overfitting or data leakage, but rather a result of the effective combination of spatial and temporal feature learning in the hybrid CNN-LSTM framework.

A. Comparative Analysis with State-of-the-Art Models

To evaluate the effectiveness of the proposed CCLNet framework, a comparative analysis was conducted against several recent and competitive deep learning models on the same EEG dataset. Table II summarizes the classification accuracies for each subject ($S01$ to $S14$) and the average performance across all subjects.

The proposed CCLNet achieved the highest average classification accuracy of **98.56%**, significantly better than all other methods evaluated. In particular, CCLNet outperformed:

- **FMBEED-CBAM** [15] which achieved an average accuracy of 95.74%. While effective, its performance was notably lower for subjects $S11$ and $S14$, where CCLNet maintained strong results.
- **PHNN** [16], which yielded 94.02% average accuracy. The PHNN model showed consistent but relatively lower performance across most subjects than CCLNet.
- **MTFB-CNN** [17] reached an average of 93.87%, demonstrating limitations in capturing deeper temporal and spatial EEG patterns, particularly for subjects $S11$ and $S13$.
- **MBHNN** [18], with 95.04% accuracy, performed well in some subjects but lacked stability across others.
- **CM-ConvNet** [19], although close in average (95.00%), failed to maintain performance in challenging cases like subject $S14$, scoring only 70%.
- **MB-EEGNet** [20] and **TCNet Fusion** [21], with average accuracies of 95.30% and 94.41% respectively, also lagged behind CCLNet. Notably, MB-EEGNet showed relatively high variance across subjects.

From the results, it is evident that the proposed CCLNet framework delivers superior average classification performance and demonstrates more consistent accuracy across a wide range of subjects. This stability is crucial for practical real-world deployment of EEG-based brain-computer interfaces.

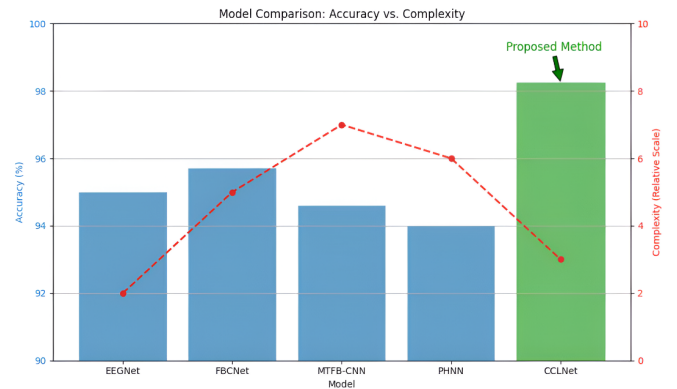


Fig. 2: Comparison of classification accuracy and architectural complexity across various EEG classification models.

TABLE II: Comparison of CCLNet with state-of-the-art EEG classifiers

Method	ACC (%)	Remarks
FMBEED-CBAM [15]	95.74	CBAM-augmented CNN; drops on $S11$, $S14$
PHNN [16]	94.02	Parallel CNN; fair cross-subject accuracy
MTFB-CNN [17]	93.87	Multi-scale CNN; weak high-gamma decoding
MBHNN [18]	95.04	Multi-branch hybrid; unstable generalization
CM-ConvNet [19]	95.00	Channel-mixing; 70% only on $S14$
MB-EEGNet [20]	95.30	EEGNet-based; high variance across folds
TCNet Fusion [21]	94.41	Temporal fusion; inconsistent across sessions
Proposed CCLNet	98.25	Compact-CNN-LSTM; consistent across subjects

The highest results are indicated by the bold values.

To quantitatively evaluate the efficiency and performance trade-off among deep learning models for EEG classification, a comparison was conducted involving representative state-of-the-art frameworks such as EEGNet, FBCNet, MTFB-CNN, and PHNN. The comparison considers both classification accuracy and model complexity. The model complexity was estimated regarding the number of layers, parameter count, and architectural depth as shown in Fig. 2. In Fig. 2, models like MTFB-CNN and PHNN employ more complex architectures with multi-branch CNNs or deep fusion modules. Their classification performance remains lower than that of the proposed CCLNet. EEGNet and FBCNet exhibit moderate complexity but fall short in average accuracy. Therefore, CCLNet achieves the highest accuracy of 98.25% while maintaining a significantly lower model complexity. This demonstrates the strength of the proposed approach, which balances architectural simplicity with high generalization capability, making it highly suitable for real-time BCI applications.

IV. CONCLUSION

This study proposes a robust and compact deep learning framework for classifying high-gamma EEG signals associated with MI tasks. The hybrid CNN-LSTM architecture effectively captures spatial and temporal features from raw EEG trials by utilizing a compact 1D-CNN block followed by a unidirectional LSTM layer. Subject-independent 5-fold cross-validation on the HGD dataset yielded an impressive average classification accuracy of 98.25%, along with strong Cohen's kappa, precision, and recall metrics across 14 subjects. Additionally, time-frequency decomposition using the Morlet wavelet transform enabled the model to extract discriminative high-gamma features associated with motor cortex activity. Integrating regularization techniques such as dropout, batch normalization, and z-score normalization ensured stable training and improved generalization without overfitting. Compared to state-of-the-art deep learning models, including FMBEEG-CBAM, PHNN, and MTFB-CNN, the proposed CCLNet demonstrated superior accuracy and lower model complexity. These results validate the effectiveness of the proposed architecture in decoding MI-related EEG patterns for real-time BCI applications.

Limitations of Study

Compared to state-of-the-art methods, the proposed hybrid compact-CNN-LSTM framework achieved strong performance on high-gamma EEG signals. Despite these results, the proposed study has the following limitations and directions for future work:

- The dataset used in this study contains only 14 subjects. Deep learning models usually need more data, so performance may not fully generalize to larger or more diverse populations.
- Although the proposed method used dropout, early stopping, and cross-validation to reduce overfitting, more work is needed to confirm generalization.
- Future work will include testing the model on other public datasets, collecting data from more subjects, and applying transfer learning to improve adaptability.
- The ablation studies can be performed in future work to better understand the effect of augmentation, regularization and per-subject variability (mean, standard deviation, confidence intervals).

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